Daniel McKenzie

Applied Math and ML.

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Positions

Colorado School of Mines. Assistant Professor — 2022–present
University of California, Los Angeles. Assistant Adjunct Professor — 2019-2022

Research interests

Optimization and signal processing applied to machine learning and data science. More specifically: zeroth-order optimization and applications, Learn-to-optimize, and data-driven metrics for unsupervised learning.

Education

University of Georgia, USA. PhD. Advisor: Ming-Jun Lai. — 2013-2019
University of Bayreuth, Germany. Short term visitor. — 03/2013-06/2013
University of Cape Town, South Africa. M.Sc (with distinction). — 2011-2013
University of Cape Town, South Africa. B.Sc (with distinction) — 2007-2010

Awards and Recognitions

Featured in Mines research magazine. — 2024
Liggett Instructor Award. — 2021
William Armor Wills Memorial Scholarship. — 2019
University Outstanding Teaching Assistant Award. — 2017
NRF Doctoral Scholarship for Study Abroad. — 2014-2016
DAAD Short Term Research Exchange (to University of Bayreuth). — 2013
DAAD-NRF Joint Masters Bursary. — 2012
UCT Council Merit Scholarship. — 2010
NRF Honours Bursary. — 2010
Jakob Burlak Memorial Trust Scholarship. — 2009
UCT Science Faculty Scholarship. — 2007,2008,2009

Publications

A list is also available online. *= undergraduate coauthor.

Journals

It begins with a boundary: A geometric view on probabilistically robust learning. ESAIM: COCV. Leon Bungert, Nicolás García Trillos, Matt Jacobs, **Daniel McKenzie**, Đorđe Nikolić, Qingsong Wang. — 2026

Fully Adaptive Zeroth-Order Method for Minimizing Functions with Compressible Gradients. Journal of Optimization Theory and Applications. Geovani Nunes Grapiglia, **Daniel McKenzie** — 2026

Curvature-Aware Derivative-Free Optimization. Journal of Scientific Computing. Bumsu Kim, **Daniel McKenzie**, HanQin Cai and Wotao Yin. — 2025

Differentiating through integer linear programs with quadratic regularization and Davis-Yin splitting. TMLR. **Daniel McKenzie**, Howard Heaton, Samy Wu Fung 2024

Normalizing Basis Functions: Approximate Stationary Models for Large Spatial Data. STAT. Antony Sikorski, **Daniel McKenzie**, Doug Nychka. 2024

Three-Operator Splitting for Learning to Predict Equilibria in Convex Games. SIMODS. **Daniel McKenzie** and Howard Heaton, Qiuwei Li, Samy Wu Fung, Stanley Osher and Wotao Yin. 2024

Fermat Distances: Metric Approximation, Spectral Convergence, and Clustering Algorithms. JMLR. Nicolás García Trillos, Anna Little, **Daniel McKenzie**, and James Murphy (alphabetical). 2024

From the simplex to the sphere: Faster constrained optimization using the Hadamard parametrization Information and Inference. Qiuwei Li, **Daniel McKenzie**, and Wotao Yin (alphabetical). 2023

A one-bit, comparison-based gradient estimator. ACHA. HanQin Cai, **Daniel McKenzie**, Wotao Yin and Zhenliang Zhang (alphabetical). 2022

Balancing geometry and density: Path distances on high-dimensional data. SIMODS. Anna Little, **Daniel McKenzie**, and James M. Murphy (alphabetical). 2021

Zeroth-order regularized optimization (ZORO): Approximately sparse gradients and adaptive sampling. SIOPT. HanQin Cai, **Daniel McKenzie**, Wotao Yin and Zhenliang Zhang (alphabetical). 2021

Compressive sensing for cut improvement and local clustering. SIMODS. Ming-Jun Lai and **Daniel McKenzie** (alphabetical). 2020

Power weighted shortest paths for clustering Euclidean data. Foundations of Data Science. **Daniel McKenzie** and Steve Damelin 2019

Conferences

LatticeVision: Image to Image Networks for Modeling Non-Stationary Spatial Data AISTATS 2026 (~25% acceptance rate). Antony Sikorski, Michael Ivanitskiy, Nathan Lenssen, Douglas Nychka, **Daniel McKenzie**. 2026

On logical extrapolation for mazes with recurrent and implicit networks. AAAI 2026 (oral presentation, top ~5% of submissions). Brandon Knutson, Amandin Rabeendran, Michael Ivanitskiy, Jordan Pettyjohn, Cecilia Deniz-Behn, Samy Wu Fung, **Daniel McKenzie** 2026

JFB: Jacobian-Free Backpropagation for Implicit Networks. AAAI 2022 (15% acceptance rate). Samy Wu Fung, Howard Heaton, Qiuwei Li, **Daniel McKenzie**, Stanley Osher and Wotao Yin. 2022

A zeroth-order, block coordinate descent algorithm for huge-scale black-box optimization. ICML 2021 (22% acceptance rate). HanQin Cai, Yuchen Lou*, **Daniel McKenzie**, and Wotao Yin (alphabetical). 2021

Who killed Lilly Kane? A case study in applying knowledge graphs to crime fiction. IEEE Big Data GTA3 Workshop. With Mariam Alaverdian*, William Gilroy*, Veronica Kirgios*, Xia Li, Carolina Matuk*, **Daniel McKenzie**, Tachin Ruangkriengsin*, P. Jeffrey Brantingham and Andrea Bertozzi (alphabetical). 2020

Submitted

Gradient-Free Optimization for Matrix Functions. Sawyer Allen*, Cash Cherry*, Aidan Eck*, Stephen Becker, **Daniel McKenzie**. arXiv:2609.03170. 2026

Interpreting Language Model Hidden States at Scale. Jordan Pettyjohn, Mansi Sakarvadia, Nathaniel Hudson, **Daniel McKenzie**, Kyle Chard, Ian Foster. arXiv:2608.10260. 2026

A Non-stationary, Amortized, Transfer Learning Approach for Modeling Italian Air Quality. Alessandro Fusta Moro, Antony Sikorski, **Daniel McKenzie**, Alessandro Fassò, Douglas Nychka. arXiv:2604.18823. 2026

Misc.

Adapting Zeroth Order Algorithms for Comparison-Based Optimization Isha Slavin* and **Daniel McKenzie** SIURO. 2023

Learning to Optimize SIAM News. Samy Wu Fung, **Daniel McKenzie**, Wotao Yin. 2022

Talks

All talks invited unless otherwise stated.

SIAM OPT. Edinburgh, UK.	06/2026
CIROH DevCon. Salt Lake City, UT, USA.	05/2026
Lehigh ISE Seminar. Bethlehem, PA, USA.	11/2025
Tufts CAM Seminar. Medford, MA, USA.	09/2025
ICCOPT. Los Angeles, USA.	07/2025
IISA Annual Meeting. Lincoln, NE, USA.	06/2025
SIAM MDS. Atlanta, USA.	10/2024
IMSI Workshop on Computational Imaging. Chicago, USA.	07/2024
LOL24 Workshop. Luminy, France	06/2024
Becker Group Seminar, CU. Boulder, USA.	03/2024
CSU IDA Seminar. Fort Collins, USA.	01/2024
U. Arizona Early Career Colloquium. Online	09/2023
SIAM OP: Advances in optimization with Machine Learning. Seattle, USA.	06/2023
NIST AI COI Seminar. Boulder, USA.	02/2023
Emory CODES Seminar. Atlanta, USA.	02/2023
UGA Applied Math Seminar. Athens, USA.	02/2023
George Mason University CMAI Colloquium. Fairfax, VA.	10/2022
SIAM MDS: Manifold Learning and Dimensionality Reduction. San Diego, USA.	09/2022
SIAM UQ22: Deep Learning for Optimization. Atlanta, USA.	03/2022
Math Machine Learning Seminar. Max Planck Institute, Germany.	03/2022
INFORMS2021: Recent Advances in Derivative-free Optimization. Anaheim, USA.	10/2021
Mathematics of Machine Learning Conference. Bielefeld University, Germany. (contributed)	08/2021
SIAM OP21: Optimization, Data Science and their Applications. Online.	07/2021
Optimal Transport and Mean Field Games Seminar. Online.	06/2021
INFORMS2020: Session on Recent Progress in Blackbox Optimization. Online.	11/2020
Tufts Math of Data Science Lecture Series. Online.	04/2020
UGA Applied Math Seminar. Athens, USA.	02/2020
SIAM South-Eastern Sectional. Knoxville, USA.	09/2019
International Conference on Computational Harmonic Analysis (ICCHA7). Nashville, USA. (contributed)	05/2018
AMS Central Sectional. Ann Arbor, USA.	10/2018

Teaching

	Math 500: Linear Spaces (graduate level), Mines	2024–present
	Math 551: Computational Linear Algebra (graduate level), Mines	2024–present
	Math 332: Linear Algebra, Mines.	2023, 2026
	Math 599: Analysis (independent study).	2023
	Math 551: Computational Linear Algebra (graduate level), Mines	2023
	Math 332: Linear Algebra, Mines.	2022
	Math 151BH: Honors Applied Numerical Methods II, UCLA. (Taught twice. I also co-developed this course).	2021–2022
	Math 151AH: Honors Applied Numerical Methods I, UCLA. (Taught twice. I also co-developed this course).	2021–2022
	Math 118: Mathematical Methods of Data Theory, UCLA. (Taught four times. I also co-developed this course).	2020–2022
	Math 170S: Statistics, UCLA.	2020
	Math 151A: Applied Numerical Methods I, UCLA.	2019
	Math 32A: Calculus III, UCLA.	2019
	Math2250: Calculus I, UGA. (taught three times).	2015–2019
	Math1113: Precalculus, UGA. (taught six times).	2014–2018

Service

Committees	TPD Hiring committee. Mines	2026
	Graduate Committee. Mines.	2024–present
	Linear Algebra Curriculum Redesign Committee. Mines.	2024
	Graduate Computing Resource Committee (chair). Mines.	2023
	Colloquium Committee. Mines.	2022–present
	CAM Curriculum Redesign Committee. Mines	2022
Organization	Sparse and Low Rank Methods for Unsupervised Learning Session at SIAM MDS.	2024
	Differentiating through fixed-points and applications Session at INFORMS IOS.	2024
	Learning to Optimize and Optimizing to Learn. Session at SIAM MDS.	2022
	Exploiting structure in zeroth-order optimization. Workshop at INFORMS2021.	2021
	ZOOM: Zeroth Order Online Meeting. (Online) mini-conference.	2020
Undergrad. Students Mentored	Anna Haemer (Mines)	2026
	Cash Cherry (Mines → UCSB), Aidan Eck (Mines), Sawyer Allen (Mines).	2024–present
	Amandin Chyba Rabeendran (Mines → NYU), Jordan Pettyjohn (Mines).	2023
	Yuchen Lou (Hong Kong Univ. → Northwestern Univ.), Isha Slavin (UCLA → NYU), Allen Zou (UCSD → Lawrence Berkeley National Lab.).	2021
	Mariam Alaverdian (Los Angeles Community College → Yale), William Gilroy (Harvey Mudd), Veronica Kirgios (Notre Dame), Carolina Matuk (Univ. Iowa), Tachin Ruangkriengsin (UCLA → Princeton), Charles Stoksik (UCLA), Chenglin Yang (UCLA → Columbia), Allen Zou (UCSD).	2020

	Lucas Connell (UGA).	2018

Graduate Students Mentored

Adam Martin (MS, Mines)	2026
Jordan Pettyjohn (MS, Mines → Chicago). Kate Raitz MS (Mines).	2024–2025
Ziyu Li (Mines), Colin Fenster (MS, Mines).	2024–present
Antony Sikorski (Mines), Brandon Knutson (Mines)	2022–present
Howard Heaton (UCLA), Bumsu Kim (UCLA).	2019–2022

Outreach

CSST Mentor. The resulting paper was presented at ICML.	2020
UCLA REU Mentor. The resulting paper was presented at IEEE Big Data 2020	2020
UGA MathCamp. I mentored a group of five high school students on the "Monster Epidemiology" project.	2018
SHAWCO. I tutored students, trained volunteers and co-led the KenSMART project.	2011–2012

Reviewing

Math Programming, Neurips (x6), TMLR (x2), AAAI (x4) AISTATS (x3).	2025
ACHA, JMLR, TMLR, AAAI (x3), SIAM SISC (x2), SIAM OPT, EJAM.	2024
AAAI (x3), Neurips (x4).	2023
SIMODS, IEEE TNNLS, Neurips (x2), ICML (x2).	2022
SODA.	2019

Skills

Proficient in Python (specifically: NumPy, SciKitLearn and PyTorch), Matlab, LaTeX, Git, and Markdown. Experienced in mentoring junior researchers, and in explaining technical concepts to non-technical audiences.